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Retention and transformation of exogenous Hg in acidic paddy soil under alternating anoxic and oxic conditions: Kinetic and mechanistic insights☆

Yufan Zhang [a,b,c,1], Xiangqin Wang [c,1], Yang Yang [c], Yingmei Huang [c], Xiaomin Li [a,b], Shiwen Hu [c], Kexue Liu [d], Yan Pang [c], Tongxu Liu [c,*], Fangbai Li [c]

[a] SCNU Environmental Research Institute, Guangdong Provincial Key Laboratory of Chemical Pollution and Environmental Safety & MOE Key Laboratory of Theoretical Chemistry of Environment, South China Normal University, Guangzhou, 510006, China
[b] School of Environment, South China Normal University, Guangzhou, 510006, China
[c] National-Regional Joint Engineering Research Center for Soil Pollution Control and Remediation in South China, Guangdong Key Laboratory of Integrated Agro-environmental Pollution Control and Management, Institute of Eco-environmental and Soil Sciences, Guangdong Academy of Sciences, Guangzhou, 510650, China
[d] School of Resources and Planning, Guangzhou Xinhua University, Guangzhou, 510310, China

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ABSTRACT

To estimate the risks and developing remediation strategies for the mercury (Hg)-contaminated soils, it is crucial to understand the mechanisms of Hg transformation and migration in the redox-changing paddy fields. In present study, a Hg-spiked acidic paddy soil (pH 4.52) was incubated under anoxic conditions for 40 d and then under oxic conditions for 20 d. During anoxic incubation, the water-soluble, exchangeable, specifically adsorbed, and fulvic acid-complexed Hg decreased sharply, whereas the humic acid-complexed Hg, organic, and sulfide-bound Hg gradually increased, which were mainly ascribed to the enhanced adsorption on the surface of soil minerals with an increase in soil pH, complexation by organic matters, precipitation as HgS, and absorption by soil colloids triggered by reductive dissolution of Fe(III) oxides. By contrast, after oxygen was introduced into the system, a gradual increase in available Hg occurred with decreasing soil pH, decomposition of organic matters and formation of Fe(III) oxides. A kinetic model was established based on the key elementary reactions to quantitatively estimate transformation processes of Hg fractions. The model matched well with the modified Tessier sequential extraction data, and suggested that large molecular organic matter and humic acid dominated Hg complexation and immobilization in acidic paddy soils. The content of methylmercury increased and reached its peak on anoxic 20 d. Sulfate-reducing bacteria *Desulfovibrio* and *Desulfomicrobium* were the major Hg methylating bacteria in the anoxic stage whereas demethylating microorganisms *Clostridium_sensu_stricto_1* and *Clostridium_sensu_stricto_12* began to grow after oxygen was introduced. These new dynamic results provided new insights into the exogenous Hg transformation processes and the model could be used to predict Hg availability in periodically flooded acidic paddy fields.

1. Introduction

The highly toxic heavy metal mercury (Hg) has attracted considerable attention because it causes global environmental and health problems (Beckers and Rinklebe, 2017; Wang et al., 2022). Mercuric Hg (Hg (II)) is the predominant species of Hg in soil and tends to be highly immobilized in surficial soil (Devai et al., 2005), in which mineral surfaces have high affinity for Hg^{2+} and functional groups such as carboxylic, hydroxyl, and S-containing ligands on organic matter (OM) facilitate stable Hg^{2+} binding (Beckers and Rinklebe, 2017; Powell et al., 2004). However, under appropriate conditions such as flooded paddy soil, inorganic Hg can be released, methylated into virulent

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* Corresponding author.
E-mail address: txliu@soil.gd.cn (T. Liu).
[1] Co-first authors: These authors contributed equally to this work.

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methylmercury (MeHg), and bio-amplified along the food chain (O'Connor et al., 2019). Higher Hg exposure risks to humans occur mainly via the consumption of rice grain from Hg contaminated paddy fields (Meng et al., 2012), which are characterized by alternating anaerobic (flooding) and aerobic (drainage) conditions. The biogeochemical properties of paddy soil, as well as the mobility and transformation processes of Hg under the two conditions, may be different (Kerin et al., 2006).

Generally, the adsorption, desorption, oxidation, reduction, methylation, and demethylation processes of Hg can be changed by altering the redox potential (Eh), pH, OM type (Frohne et al., 2012), soil grain size distribution, the transformation of Fe, Mn, N, and S mediated by microbial activities (Qu et al., 2019). Under flooding conditions, with a decrease in Eh, the pH increases and may facilitate Hg adsorption to soil minerals or interaction with dissolved OM to form solid HgS (Skyllberg and Drott, 2010). However, hydrogen ions (H^+) in acidic soils have been reported to compete with Hg for active sorption sites on negatively charged mineral surfaces, releasing Hg into the liquid phase under both anaerobic and aerobic conditions (Shen et al., 2021). Once dissolved OM (e.g., fulvic acid and thiol ligands) was present, the insoluble HgS may be dissolved due to the formation of soluble complexes (Jacobson et al., 2005). Moreover, the reductive dissolution of iron (Fe) and manganese (Mn) oxides leads to the dissolution of their adsorption or co-precipitation of Hg (Ullrich et al., 2001). Additionally, the chemical reduction of Hg^{2+} and microbial-mediated MeHg formation or degradation (also called MeHg demethylation) can occur under anaerobic conditions (Barkay and Gu, 2021; O'Connor et al., 2019). Humic substances, such as humic acid, can reduce Hg^{2+} to Hg^0, and the speed of the process increases at low pH ($\leq$4) (Chakraborty et al., 2015). Microorganisms such as methanogens, iron-reducing bacteria (FeRB), and sulfate-reducing bacteria (SRB) mediate the methylation of Hg in paddy fields (Kerin et al., 2006; Tang et al., 2020). Mobile humic- or fluvic-complexed Hg was susceptible to Hg methylation, in which NH_4^+, Fe^{2+}, Mn^{2+}, and S^{2-} are electron donors, whereas O_2, NO_3^-, MnO_2, FeOOH, and SO_4^{2-} are electron acceptors. Notably, the abundance of OM and anions, such as NO_3^- in paddy soil, is also beneficial to MeHg demethylation (Helmrich et al., 2022), which is usually promoted by microbial oxidation under anaerobic conditions (Bridou et al., 2011; Oremland et al., 1991) or microbial reduction under aerobic conditions (Hu et al., 2013). Methyl trophic bacteria, methanogens, and SRB may also promote oxidative Hg demethylation under anaerobic conditions (Marvin-DiPasquale and Oremland, 1998). However, it is still unclear to quantify the contributors dominating the transformation of Hg under alternating redox conditions. Furthermore, it is still lacking a comprehensive understanding of Hg transformation affected by the variation of paddy soil properties. Therefore, it is quite essential to identify the key factors controlling Hg behavior and accurately predict the fate of Hg in paddy fields under periodic irrigation is necessary.

Regarding the unique volatility and methylation of Hg, attempts to study its dynamics and potential availability have been challenging (Huang et al., 2020). Although the mobility of Hg is related to various solids, such as OM, oxides, and hydroxides of Fe and Mn (Bourg and Loch, 1995), there is no unified standard for the classification of soil Hg fractions. Sequential extraction procedures (SEPs) have also been applied to quantify soil Hg speciation based on low, medium, or high mobility (Dong et al., 2019; Tessier et al., 1979). Although the reagents used in SEPs affected the speciation of Hg and the accuracy of derived speciation data (Dong et al., 2019; O'Connor et al., 2019), the SEPs is still helpful and necessary to estimate the speciation of Hg in soil (Frohne and Rinklebe, 2013). Therefore, further research should be conducted to accurately determine Hg speciation and dynamics in periodically flooded paddy soils.

The objectives of this study were to (1) investigate the mobility and methylation kinetics of exogenous Hg in acidic paddy soils under flooding-drainage redox conditions; (2) elucidate the key factors and mechanisms dominating Hg transformation; and (3) establish a corresponding kinetic model to clarify the processes of exogenous Hg mobility, methylation, and demethylation in acid paddy soil.

2. Materials and methods

2.1. Soil sampling and characterization

Acid surface soil (depth 0–20 cm) was collected from a non-Hg-contaminated paddy field in Yingtan City (28°13′45″N, 16°54′38″E), China. The soil sample was air-dried, homogenized, and passed through a 0.15 mm sieve to analyze the physicochemical properties according to the relevant standard methods. Soil pH (4.52), cation exchange capacity (CEC), OM, the content of carbonate, clay, Crystalline Fe and Mn oxides (DCB-Fe and DCB-Mn), amorphous Fe and Mn oxides (Ox-Fe and Ox-Mn), total Fe, Mn and Hg were determined using the methods presented in Text 1 of the supporting information (SI). The soil physical and chemical properties are listed in Table S1.

2.2. Soil incubation experiments

The changing redox conditions during flooding and drainage in paddy fields were simulated with a biogeochemical microcosm system developed by Fulda et al. (2013) and Rinklebe et al. (2016), in which the oxic or anoxic conditions could be well defined and readily changed (Yang et al., 2021). The system was established in 50 mL serum bottles containing 6 g dry soil and spiked with 30 mL 4.93 mg L^{-1} $HgCl_2$ solution to obtain a final concentration of 24.65 mg kg^{-1} Hg, the relative high concentration of which was chosen for an extensive investigation of Hg transformation in contaminated paddy soil under anoxic-oxic alternated conditions. $HgCl_2$ has higher water solubility than other mercuric salts such as $Hg(NO_3)_2$ (O'Connor et al., 2019). In typical non-saturated soil conditions, $HgCl_2$ is one of the typically forms of Hg^{2+} in soil solution (Powell et al., 2004) and is much easier methylated into MeHg by microorganism compared with Hg-DOM, α-HgS and nano-Hg (Tang et al., 2020). Therefore, $HgCl_2$ was chose as exogenous Hg. All bottles were flushed with a N_2 gas stream in an O_2-free glovebox (PLAS-LABS, USA, H_2/N_2 (2/98, v/v)) and then sealed with butyl rubber stoppers and aluminum covers. The sealed bottles were taken out from the O_2-free glovebox and shaken on a horizontal shaker with a constant speed of 220 r min^{-1} at 28 °C for 40 d (anoxic period). Subsequently, the butyl rubber stoppers and aluminum covers of the bottles were replaced with a porous film (0.45 μm) to allow gas exchange with the air for 20 d with continuous shaking (oxic period).

2.3. Sample analysis

During the anoxic (0, 2, 5, 10, 20, and 40 d) and oxic (2, 7, 15, and 20 d) periods, three repeated destructive samples were collected from the incubated bottles in an O_2-free glovebox. After redox potential (Eh) and soil pH were determined by Eh and pH meter (Mettler Toledo, Switzerland), respectively, the samples were transferred into 50 mL centrifuge tubes and centrifuged at 5000 rpm for 10 min, and the supernatants were filtered through 0.45 μm filter heads (syringe) to obtain the filtrate and wet soil sample, respectively. The filtrate was divided into seven subsamples for NO_3^-, SO_4^{2-}, NH_4^+, dissolved OM (DOC), Fe(II), water-soluble Hg (F0–Hg), and water-soluble MeHg (WMeHg) analysis. A wet soil sample was used to determine total organic carbon (TOC), Fe, Hg, and MeHg. Subsamples (wet soil) used for available Fe analysis were extracted using 0.5 mol L^{-1} HCl. The subsamples used for total Hg and MeHg analyses were acidified with 5‰ HCl for further analysis.

The concentrations of NO_3^- and SO_4^{2-} were measured by ion chromatography (ICS–600, Dionex, U.S.A.). NH_4^+ was quantified using a continuous flow analyzer (SAN++, Skalar) (Cheng et al., 2017). Total organic carbon (TOC) and dissolved organic carbon (DOC) and were measured using a total organic carbon analyzer (TOC-L-CPH, Shimadzu, Japan). Concentrations of dissolved Fe(II) and HCl-extractable Fe(II) (Fe

$(II)_{HCl}$) were determined using the o-phenanthroline colorimetric method with a UV–vis spectrophotometer (TU–1950, Persee, China). The concentration of HCl-extractable total Fe (T-Fe_{HCl}) was analyzed using inductively coupled plasma-optical emission spectrometry (ICP-OES, Optima 8000, PerkinElmer, USA). The concentration of Fe $(III)_{HCl}$ was calculated as T-Fe_{HCl} minus that of $Fe(II)_{HCl}$.

A modified Tessier SEP (Tessier et al., 1979) was used to investigate Hg speciation during the incubation period. In brief, exchangeable Hg (F1), specifically adsorbed Hg (F2), fulvic acid complexed Hg (F3), humic acid complexed Hg (F4), Fe–Mn oxides bound Hg (F5), organic- and sulfide-bound Hg (F6), and residual Hg (F7) were obtained according to the SEP method provided in Table S2. The residual Hg and total Hg (THg) in the soil were digested using aqua regia in a 98 °C water bath. The F0–Hg in the filtrate, F1–Hg to F6–Hg in the extracted solutions, and F7–Hg and total Hg in the digested solutions were determined using an atomic fluorescence spectrometer (AFS-933, Jitian Instruments Co., Ltd, Beijing, China), the detailed methods are presented in Text 2 of SI. Using the modified Tessier SEP, the recovery of Hg, that is, the sum of the fractions relative to THg, varied between 91% and 108%, indicating the effectiveness of this method. Three MeHg species, WMeHg in filtrate subsamples, effective MeHg (EMeHg), and total MeHg (TMeHg) in wet soil subsamples were determined. The filtrate subsamples were distilled to concentrate water-soluble MeHg (WMeHg) (Jiang et al., 2004). Each wet soil subsample for MeHg analysis was freeze dried firstly and then divided into two parts. One part was subjected to sequential extraction with 20% $CuSO_4$–HNO_3 and CH_2Cl and water back-extraction with heating to determine TMeHg (He et al., 2004). The other part was extracted using $(NH_4)_2S_2O_3$ to determine EMeHg, which has been deemed as the bioavailable MeHg in soil (Zhu et al., 2015). All MeHg extractions were ethylated into methylethyl-Hg for analysis by cold vapor atomic fluorescence spectroscopy (CVAFS) (MERX, Brooks Rand Instruments, Seattle, WA, U.S.A.) according to the USEPA method 1630. Standard materials GBW07453(GSS-24) and ERM-cc580 (estuarine sediment) for soil THg and MeHg analysis, respectively, were used for quality control. Analysis of the standard material yielded THg and MeHg concentrations of 70 ± 8 μg kg^{-1} and 83 ± 5.7 μg kg^{-1}, respectively, which were close to the certified values of 75 ± 7 μg kg^{-1} and 75.5 ± 4 μg kg^{-1}, respectively.

The soil in the tube was used to analyze the physical characteristics of the soil and microbial communities. For the soil colloids, the samples in anoxic 0, 40 d, and oxic 15 d were centrifuged at 5000 rmp for 15 min, and then the unfiltered supernatants were scanned using a UV–vis spectrophotometer (TU-1950, Persee, China) (Berho et al., 2004). The surface site concentrations (*Hs*) were obtained with continued potentiometric titrations as well as a Gran plot, the methods were presented in the Text 3 in SI. The distribution of elements (O, C, Al, and Fe) during the anoxic-oxic period were scanned using a transmission electron microscopy (TEM, Talos f200x, U.S.A.). Frozen soil samples at anoxic 0, 20, and 40 d and oxic for 15 d were used to investigate the bacterial community using 16 S rRNA gene high-throughput sequencing. In detail, 0.5 g of wet soils (three replicates) were weighed and utilized for DNA extraction. Extraction of soil DNA was performed using a soil microbe DNA kit (MOBIO/Qiagen USA) by following the instructions provided by the manufacturer. v3 and v4 of 16 S rRNA were amplified, and DNA sequencing was performed using the Illumina MiSeq platform. High-quality sequences were clustered into operational taxonomic units (OTUs) at an identity level of 97%, and bacterial taxonomy was assigned using the Ribosomal Database Project (RDP) classifier after filtering low-quality or ambiguous reads and removing chimeric sequences. Information analysis comprised division features (OTUs and ASVs), difference analysis, diversity analysis, correlation analysis, and function prediction analysis.

2.4. Statistical analysis and modelling approaches

All data are presented as the mean ± standard error (n = 3). Microsoft Office Excel 2019 software and Origin 2021 were used for data processing and mapping. The kinetic model was further fitted according to the experimental data and reaction equations using the program Kintek Explorer 8.0 (Johnson et al., 2009). The best values of the rate constants were obtained using the chi^2 test according to our previous work (Chen et al., 2021).

3. Results and discussion

3.1. Kinetic characteristics of Hg transformation

3.1.1. Kinetics of Hg fraction transformation

Exogenous Hg(II) undergoes partitioning, transformation, and equilibration processes in soil (Tang et al., 2020). Changes in extractable fractions of Hg are depicted in Fig. 1. Generally, the relatively immobile Hg fractions, that is, the organic- and sulfide-bound Hg (F6), residual Hg (F7), and humic acid complexed Hg (F4), dominated and accounted for more than 73%, 10%, and 6% of total Hg, respectively, in all the samples (Fig. 1a). F4–Hg is insoluble and unavailable because of its covalent combination with the functional groups of humic acid covalently and the formation of gridding chelates (Wang et al., 1997). Except those on the 0 d, the amounts of F4–Hg in soil are 2 orders of magnitude higher than those of F3–Hg (fluvic acid complexed Hg). Therefore, the humic acid in soil played an important role in immobilizing exogenous Hg(II), and effectively reduced the availability of Hg in soil (Huang et al., 2012). F6–Hg represents Hg complexing with larger molecular weight OM or binding on matrix surfaces (O'Connor et al., 2019) in paddy soils, which are usually rich in OM with SH- groups (Reddy and DeLaune, 2008). This result is consistent with the literature that Hg speciation in contaminated soils and sediments is mainly determined by several variables, such as OM and sulfide (Frohne and Rinklebe, 2013; Liu et al., 2022). The Hg residual fraction (around 10%) might contain Hg bound on the surfaces of crystalline Fe (hydr)oxides, which can be reduced under strong reducing-conditions (Patrick and Jugsujinda, 1992), whereas cannot be dissolved easily in the NH_2OH–HCl solution (Fortin et al., 1993; La Force et al., 2000). With the sharp increase of soil pH under anoxic condition, the negative charges on the crystalline Fe (hydr)oxides were enhanced as well as large quantities of Hg were bound on the mineral surfaces. During the equilibration phase (0 d), the dissolved Hg (F0) and fluvic acid complexed Hg (F3) accounted for much larger percentages (4.1% and 6.7%, respectively) than specifically adsorbed Hg (F2), Fe–Mn oxides bound Hg (F5), and exchangeable Hg (F1), similar to the results of another study (Huang et al., 2012). The initial available Hg was higher on the 0 d than on the other days, which could be attributed to the large amounts of spiked mobile Hg into the system. F3–Hg is also readily available for rice (Frohne and Rinklebe, 2013) because of its association with the functional groups of fulvic acid by electrostatic forces over a wide pH range (Boszke et al., 2003). The Hg availability on 0 d was higher than that on the other days, because the large amounts of mobile Hg were spiked into the system. Compared with those on 0 d, the available F0–Hg, F1–Hg, F2–Hg, and F3–Hg sharply decreased ($p < 0.01$) to nearly undetectable levels on 2 d and 5 d and then increased slightly from 10 to 40 d, whereas the F4–Hg and F6–Hg increased significantly ($p < 0.05$), probably due to the faster adsorption or co-precipitation of Hg on soil minerals. It has been reported that the binding intensity of Hg on soil surfaces increased with anoxic incubation (Han et al., 2006). By contrast, once O_2 was introduced into the system, a gradual increase in F0–Hg ($p < 0.01$) occurred from 42 to 60 d, indicating that part of the binding of Hg on soil minerals was released into the soil solution. The degradation of OM or oxidization of HgS under oxic conditions can result in the release of organic and sulfide-bound Hg (Holley et al., 2007).

3.1.2. Kinetics of MeHg transformation

The changes in WMeHg, TMeHg, and EMeHg exhibited similar linear tendencies (Fig. 1b), characterized by an initial increase, reaching peak

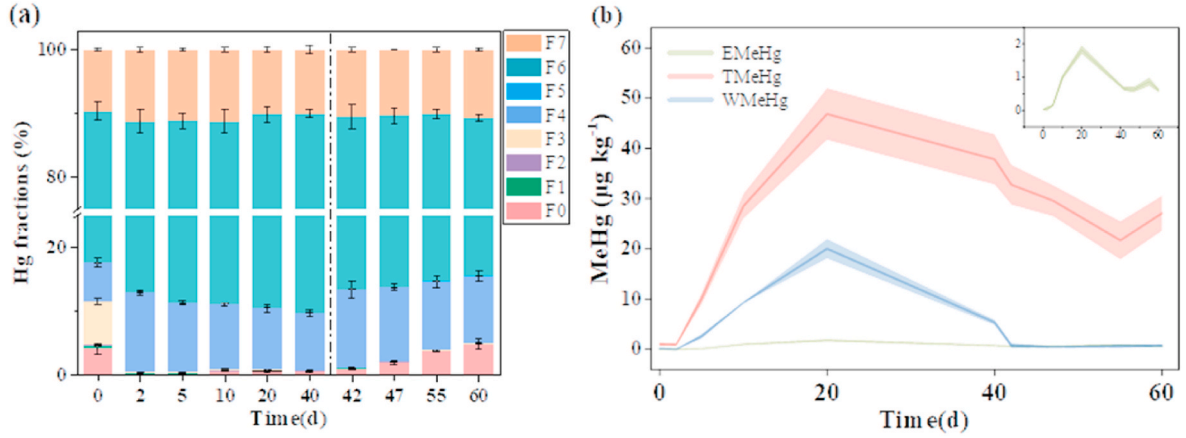


Fig. 1. Changes in Hg fractions (Water-soluble Hg (F0), exchangeable Hg (F1), specifically adsorbed Hg (F2), fulvic complex Hg (F3), humic complex Hg(F4), Fe–Mn oxides bound Hg (F5), organic and sulfide bound Hg (F6), and residual Hg (F7)) determined by sequential chemical extraction (a) and MeHg concentration extracted by three methods (water-soluble MeHg (WMeHg), effective MeHg (EMeHg) and total MeHg (TMeHg)) (b) within the 40 d anaerobic stage and 20 d aerobic stage.

values on 20 d and then decreasing gradually. The concentrations of WMeHg, TMeHg, and EMeHg were 20.03 μg kg^{-1}, 46.85 μg kg^{-1}, and 1.82 μg kg^{-1}, respectively, on 20 d. Hg(II) methylation facilitated by microorganisms produce MeHg under anoxic conditions. Biological methylation of exogenous Hg(II) with high activity in soil was more rapid than that of indigenous Hg(II), so the enhancement of the Hg methylation rate and the increase in MeHg from 0 to 20 d should be ascribed to the vigorous metabolic capability of Hg methylating microorganisms and the large amount of spiked free Hg(II) in soil solution. In the late period of anoxic incubation (20–40 d), with the consumption of electron accepters such as Fe(III) and SO_4^{2-} (Fig. 2b and c), the incorporation of free Hg(II) in soil particles such as large molecular OM, humic acid, S, and other minerals, the richness and diversity of Hg methylating microorganisms decreased (Fig. 4a), and finally the MeHg concentrations from methylation in soil began to decrease. Notably, the accumulation of MeHg in soil is a dynamic balance between inorganic Hg methylation and MeHg demethylation (Tang et al., 2020; Zhao et al., 2020). Oxidative demethylation of MeHg may occur, and the rate may be greater than the methylation rate in the late anoxic stage (Lu et al., 2016; Zhao et al., 2020). The introduction of O_2 accelerated the decrease in MeHg concentration (Fig. 1b). The concentration of WMeHg sharply decreased from 5.35 μg kg^{-1} on anoxic 40 d to 0.78 μg kg^{-1} on oxic 42 d ($p < 0.01$). TMeHg and EMeHg ranged from 21.69 to 32.74 μg kg^{-1} and 0.60–0.86 μg kg^{-1} in the oxic period, respectively, which were nearly half of the maximum value at 20 d. These results indicate that anoxic conditions were more conducive to Hg methylation than oxic conditions. The extremely low concentration of WMeHg observed from 42 to 60 d might be attributed to microbial reductive demethylation and the sorption and coprecipitation of MeHg with solid phases of soil (Zhang et al., 2010).

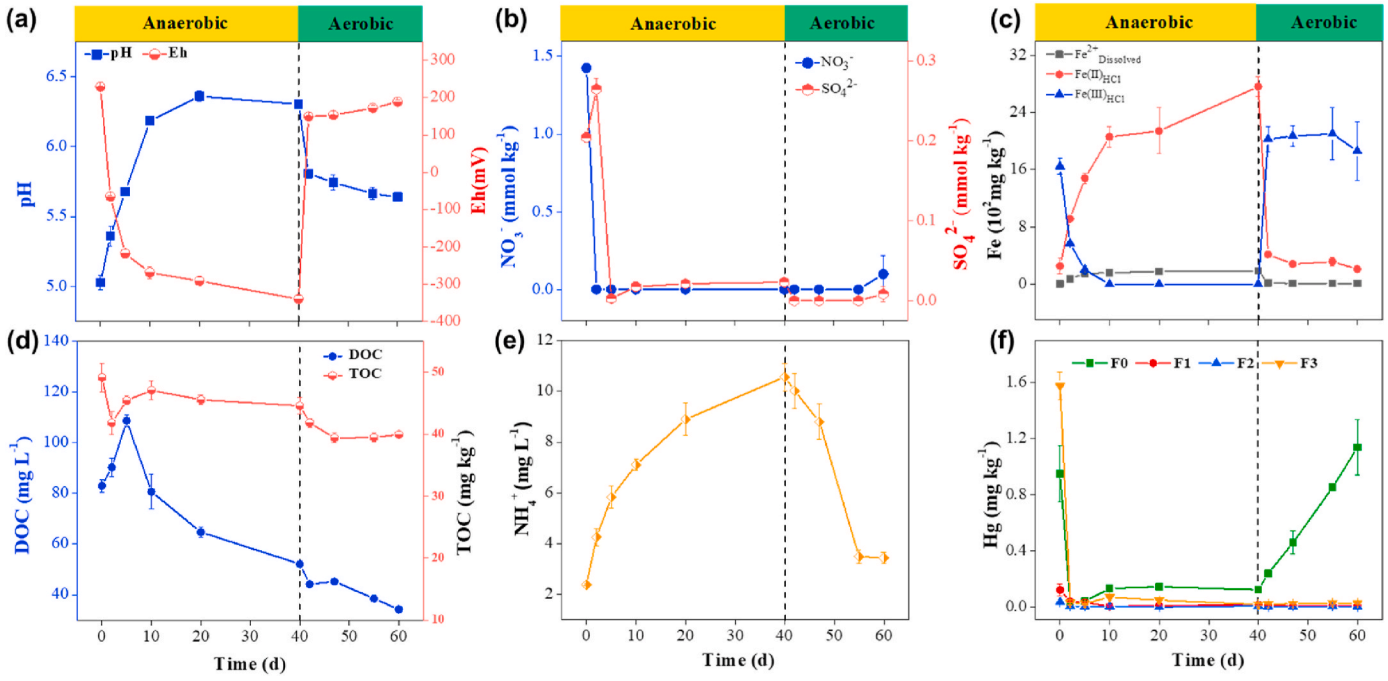


Fig. 2. Changes in soil pH and Eh (a), concentrations of SO_4^{2-} and NO_3^- (b), concentrations of dissolved Fe^{2+} and Fe fractions extracted by HCl (c), concentrations of dissolved organic carbon (DOC) (d), concentration of NH_4^+ (e), and concentrations of available Hg (f) under anaerobic and aerobic conditions.

3.2. Changes in chemical characteristics

The changing tendencies of Eh and pH values with time in the acidic paddy soil were opposite (Fig. 2a), which was consistent with Beckers et al. (2019a). Under anoxic conditions (0–40 d), a strong reducing condition was established in the first 10 d, with Eh decreasing rapidly from 229 mV to −269 mV, and the pH increased rapidly from 5.0 to 6.18. The reverse relationship of Eh and pH was presumably attributed to the consumption of protons required for reducing oxidants such as Fe^{3+}, SO_4^{2-} and NO_3^- (Beckers et al., 2019a, 2019b). The increase in pH in acidic soils depends on the biological reduction of oxidants and the consumption of protons under anaerobic conditions (Reddy and DeLaune, 2008). As shown in Fig. 2b, c, and 2e, concentrations of Fe $(III)_{HCl}$, NO_3^-, and SO_4^{2-} sharply decreased ($p < 0.01$), whereas those of $Fe(II)_{dissolved}$, $Fe(II)_{HCl}$, and NH_4^+ gradually increased during the anoxic period ($p < 0.01$), reflecting the onset of microbial reduction of Fe(III) (hydr)oxides NO_3^- and SO_4^{2-}. Moreover, the concentrations of DOC increased within the first 5 d and then decreased gradually ($p < 0.05$) (Fig. 2d). Moreover, the concentrations of DOC increased within the first 5 d and then decreased gradually ($p < 0.05$) (Fig. 2d), which might be attributed to the release of low molecular OM from soil minerals first with an increase in pH (Grybos et al., 2009; Yang et al., 2021), followed by consumption by microorganisms or Hg complexation processes under anoxic conditions (Graham et al., 2012; Gu et al., 2011; Li and Gong, 2021). For example, as one of the most important soluble OMs, humic acid can combine with Hg^{2+} to form insoluble complexes at the soil-water interface (Chadwick et al., 2006; Mason et al., 2006). Compared with DOC, the concentration of TOC fluctuated slightly during the entire incubation period. After O_2 was introduced into the system, Eh was enhanced immediately (approximately 400 mV, $p < 0.01$) and was close to that at 0 d (229 mV), whereas pH decreased immediately (approximately 0.5) within the first 2 d. Contrary to that under anoxic conditions, the pH gradually decreased under oxic conditions ($p < 0.05$), and it should be controlled by the oxidation of reductants (such as Fe(II), NH_4^+, and S^{2-}) and the release of protons (Kögel-Knabner et al., 2010), because the $Fe(III)_{HCl}$ increased immediately ($p < 0.01$) but the dissolved Fe^{2+}, $Fe(II)_{HCl}$, and NH_4^+ decreases sharply (Fig. 2c and e). Subsequently, the decrease in DOC was further enhanced by sorption or coprecipitation with newly formed Fe oxides (Fig. 2d).

The mobility and transformation of Hg are affected by variations in soil properties (Frohne et al., 2012). Under anoxic conditions, the decrease in available Hg can be ascribed to the following reasons: (i) the increase in pH enhances the negative charges on soil minerals, decreasing the solubility of Hg by promoting its adsorption on the surfaces of these minerals (Naidu et al., 1997). As shown in Fig. 2a and f, the changes in pH and dissolved Hg (F0) were almost opposite, indicating substantial adsorption of dissolved Hg onto soil minerals with an increase in pH. A similar behavior was generally observed by Frohne et al. (2012). (ii) The mobilization of Hg was inhibited by its complexation with OM, such as humic acid and larger molecular weight OM (Beckers and Rinklebe, 2017), whereas soluble Hg complexes associated with dissolved OM, such as fluvic acid, may be reduced (Skyllberg and Drott, 2010), which can be confirmed by the decrease in F3–Hg and DOM and the increase in F4–Hg and F6–Hg. (iii) The activities of various microorganisms, such as SRB and FeRB, were stimulated by minimizing the exposure to air. As mentioned in other sections, SRB and FeRB can potentially involve inorganic Hg methylation using SO_4^{2-} and Fe(III) as electron acceptors and produce sulfide (S^{2-}) and Fe(II) (Gilmour et al., 1992), respectively. S^{2-} can react with Hg^{2+} to form insoluble HgS, effectively inhibiting MeHg production under reduction conditions (Lei et al., 2021). In line with the rapid depletion of SO_4^{2-} and generation of Fe(II), the formation of MeHg first increased rapidly but then decreased in the late anoxic period (Fig. 1b), whereas F6–Hg increased with anoxic incubation ($p < 0.05$) (Fig. 1a). Increasing concentrations of MeHg occurred concomitantly with decreasing SO_4^{2-} concentrations (Beckers et al., 2019a). Some other anaerobic microorganisms, such as nitrate-dependent Fe(II)-oxidizing bacteria (FeOB), can oxidize Fe(II) to biogenic Fe(III) oxides, which have a high sorption capacity for Hg or the ability for co-precipitating with Hg compounds (Vasques et al., 2020). However, with the rapid depletion of NO_3^- in the first 2 d, the anoxic Fe(II) oxidation should be very weak, as the proportion of Fe–Mn oxide-bound Hg (F5–Hg) in all the samples was negligible (less than 0.16%) during the anoxic incubation (Fig. 1a). Determining whether the microbial-mediated reduction of SO_4^{2-}or reductive dissolution of Fe oxides plays a substantial role in the mobility and methylation of Hg (Mason et al., 2006). Hg is more easily bound to OM and sulfide fractions than to Fe oxide fractions. However, biogeochemical cycling of iron and sulfur usually co-occur or interact with each other (Hellal et al., 2015). Inorganic sulfur mainly exists as HS^-, S^{2-}, or S^0, allowing the ready formation of FeS or FeS_2 under the anoxic condition, influencing the mobility and methylation of Hg via co-precipitation or adsorption processes (Zhao et al., 2016).

Once O_2 was introduced into the system, the bounce back of Eh would induce the decomposition of OM and oxidative dissolution of HgS, and a rapid decrease in pH would be favorable for Hg desorption from the surface of soil minerals. Consequently, a gradual decline in F6–Hg (Fig. 1a) and a steady increase in F0–Hg (Fig. 2f) occurred. According to the Hg speciation in the Eh-pH diagram, Hg^{2+} is the dominant species with high solubility under aerobic and acidic soil conditions (Reddy and DeLaune, 2008).

3.3. Changes in physical characterization

The UV–vis spectra of the soil suspensions collected under anoxic 0 and 40 d and oxic 20 d, are shown in Fig. 3b. The spectra of all three suspensions decreased from the UV to the visible region, and the absorbance value of the suspension on anoxic 40 d was the highest, followed by those on oxic 60 d and anoxic 0 d. The soil suspension is mainly composed of suspended solids and colloids, which have different responses in the UV–vis wavelength range. Suspended solids and colloids mainly contribute to the range of 200–300 nm and 450–800 nm, respectively, and those in the range 300–450 nm are equal for colloid and suspended solids (Berho et al., 2004). Therefore, the results suggest that colloid amounts are released under anoxic conditions and re-immobilized under oxic conditions.

The release and adsorption of colloids are dependent on the transformation of soil minerals such as Fe oxides, OM, and clay minerals (Tombácz et al., 2004). The morphology of soil particles and distribution of C, O, Al, and Fe in the soil suspension collected under anoxic (0 and 40 d) and oxic (20 d) conditions were detected (Fig. 3a). The HAADF images confirmed the formation of aggregates during the initial anoxic incubation. Subsequently, the aggregates increase in dispersiveness with cloud-like structures on anoxic 40 d, whereas they re-aggregated on oxic 20 d. The Fe signal in the EDS mapping correlated well with the HAADF images, indicating that the reductive dissolution of Fe oxides under anoxic conditions might cause colloid release, and Fe(II) oxidation under oxic conditions might result in the re-precipitation of colloids with the newly formed Fe oxides (Yang et al., 2021). Periodic flooding in paddy fields has been reported to trigger the release of colloids via the reductive dissolution of Fe oxides (Huang et al., 2018). The large amounts of newly released colloids under anoxic conditions expose highly specific fresh surfaces and reaction activities for Hg sorption and carry Hg within soil pores (O'Connor et al., 2019).

Fig. 3c shows that the surface site concentration (*Hs*) of the soil samples gradually increased from 15 to 149 mmol kg^{-1} during the anoxic phase, then rapidly decreased to 111 mmol kg^{-1} and stabilized at approximately 120 mmol kg^{-1} after O_2 was introduced into the system, which was similar to the change tendency of pH. *Hs* is a crucial index for

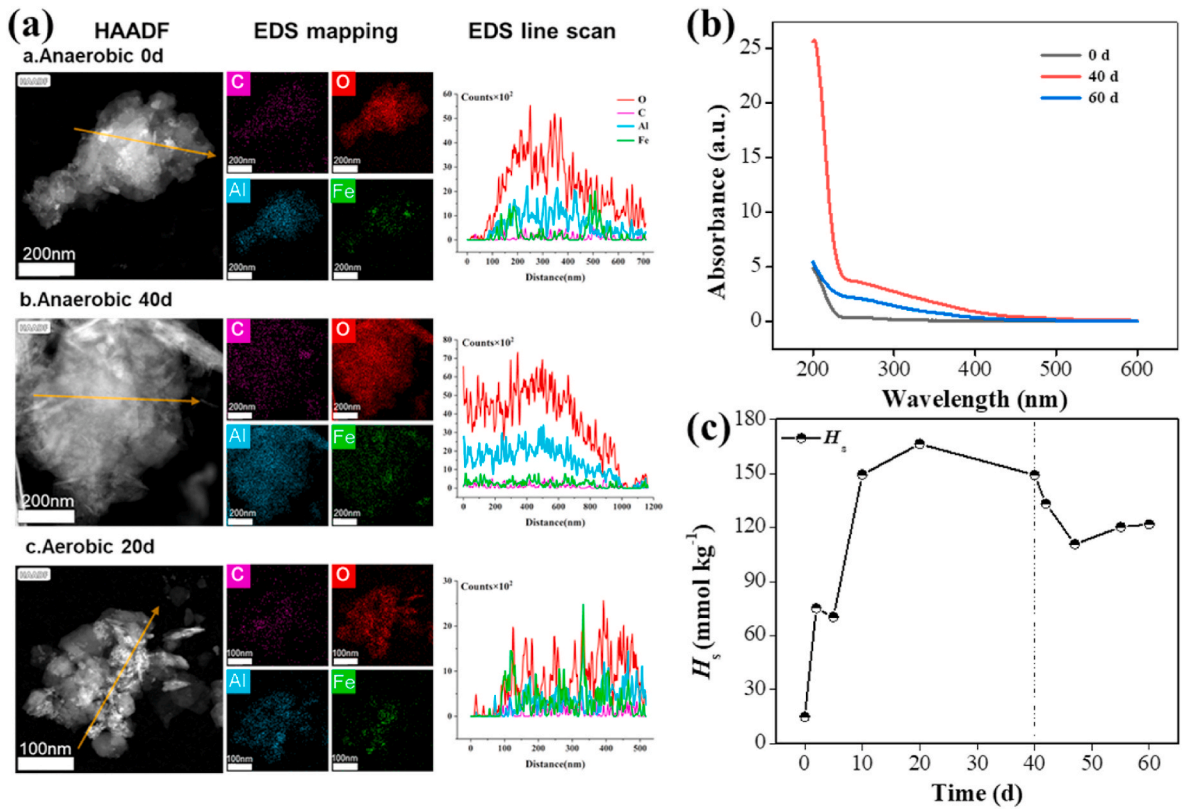


Fig. 3. High-resolution HAADF-STEM images and EDS line scans of C, O, Al, and Fe distribution (a), UV–vis spectra of soil suspensions between 200 and 600 nm (b), and the surface site concentrations (*Hs*) of soil samples under anoxic and oxic conditions.

judging the acid-base buffer capacity of variable charge components in soil because it determines the adsorption capacity of H^+ and OH^- on the surface of variable charge components (Wang et al., 2018). Therefore, an increase in *Hs* under anoxic conditions might enhance the association between Hg and soil.

3.4. Changes in composition of the microbial community

Biotic and abiotic pathways but mainly microorganisms (biotic pathway) mediated Hg methylation (Monperrus et al., 2007). From four groups of soil samples collected on 0, 20, 40, and 60 d (each group with three replicates, $3 \times 4 = 12$ soil samples), a total of 959,192 valid bacterial sequences were obtained by high-throughput sequencing of the 16 S rRNA gene. Alpha diversity, measured as the number of observed OTUs, varied between samples (262-1415 OTUs). The Chao 1 index and Shannon indices reflect the richness and diversity of species (Xu et al., 2021). Both indices showed significant ($p < 0.01$) decreasing trends with incubation time under anoxic conditions, and the Chao1 index recovered when O_2 was reintroduced into the system (Fig. 4a), indicating that continued anaerobic incubation can reduce the richness and diversity of soil bacteria (Beckers et al., 2019a). The differences in the composition of microbial communities between different samples are usually compared with beta-diversity (Li et al., 2020), which was analyzed with principal coordinates analysis (PCoA) based on the binary Jaccard algorithm (Fig. 4b). PC1 and PC2 contributed to 15.91% and 13.85% of the variance, respectively. The distance between the groups on 20 d and 40 d was very close, both of which were significantly separated from the initial anoxic 0 d and 60 d groups, indicating a considerable discrepancy in microbial communities under oxic and anoxic conditions.

The microbial community at the phylum level (Fig. 4c) revealed that *Firmicutes* (26–86%), *Proteobacteria* (1–44%), *Acidobacteriota* (6–17%), and *Chloroflexi* (1–9%) were the dominant bacteria. The relative abundance of *Firmicutes* substantially increased with anoxic incubation time but decreased sharply under oxic conditions. The relative abundance of *Proteobacteria* on 60 d was much higher than that on the other three anoxic days. The relative abundance of *Bacteroidetes* presented a larger portion on 20 d than on the other days. *Proteobacteria*, *Firmicutes*, and *Bacteroidetes* have been widely reported to be Hg-resistant bacterial phyla harboring *merA/merB* genes, and their populations increase under Hg exposure (Hu et al., 2019; Liu et al., 2014). Microbes primarily responsible for MeHg production in nature are in anaerobic sediments (Gilmour et al., 2013), soils, and bottom waters. *Firmicutes* dominate some anaerobic habitats where MeHg is produced (Morel, 2016). At the genus level, high relative abundances of *Desulfovibrio* and *Desulfomicrobium* were found on 20 d, which are SRB validated for their high ability to methylate Hg (Liu et al., 2011; Frohne et al., 2012). However, methylation of Hg is strain-specific rather than genus- or species-dependent (Bridou et al., 2011). SRB community analysis at the genus level is not the only tool used to explain or predict Hg methylation in natural ecosystems (Ranchou-Peyruse et al., 2009). The results are consistent with those of Beckers et al. (2019b) that Hg methylation was the main process leading to substantial increases in MeHg under flooding conditions rather than release from the solid phase. *Clostridiumsensustricto_1* and *Clostridiumsensustricto_12* were detected in all four soil groups. These two genera usually have strong correlations with MeHg demethylation (Zhou et al., 2020) and are present at high relative abundances in soil during redox alternation (Wang et al., 2021). Their relative abundance decreased to the lowest value on 20 d, then increased on 40 d, but decreased again on 60 d, indicating the weakest biological demethylation of Hg and corresponding to the highest production of MeHg on 20 d.

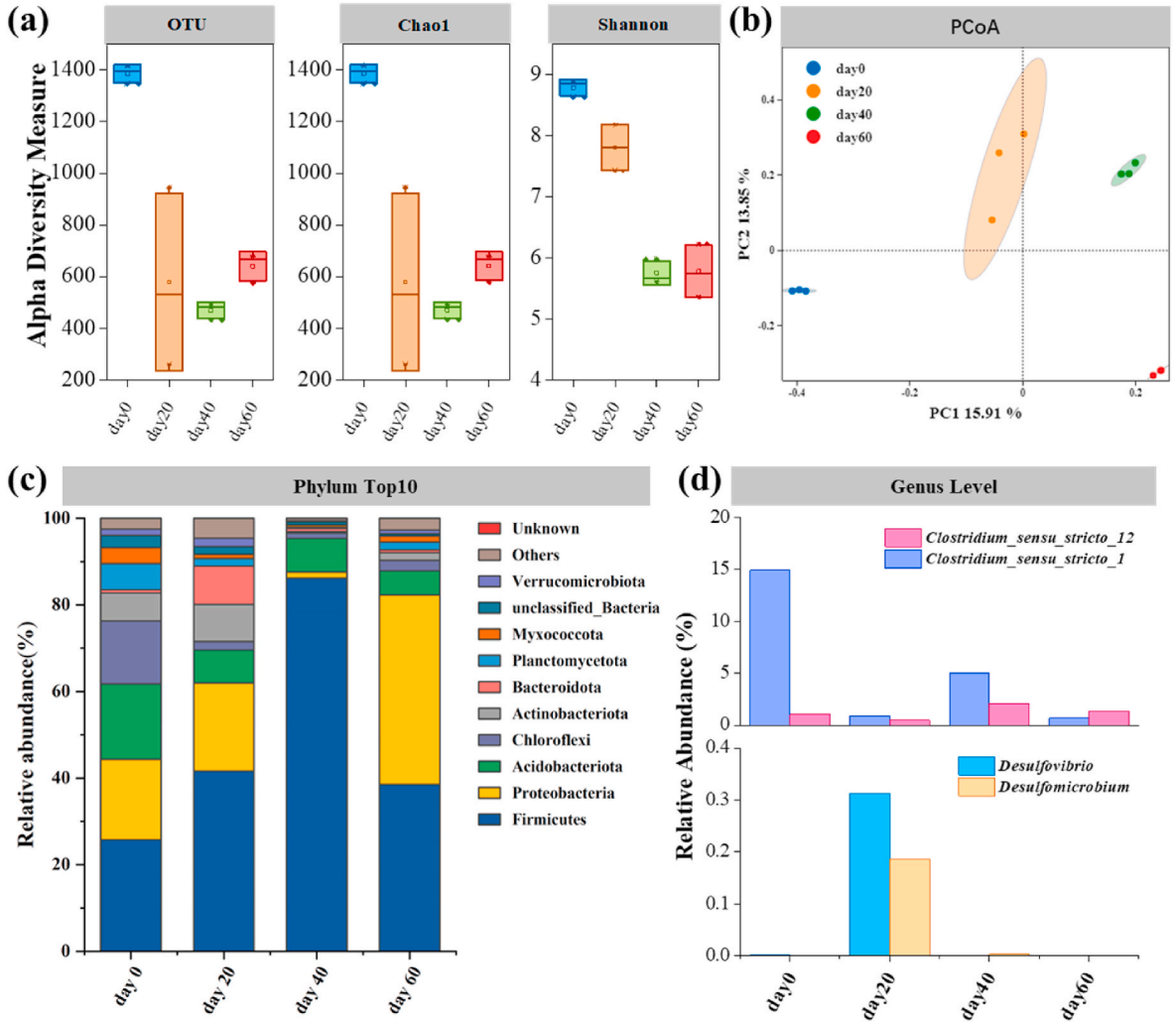


Fig. 4. Bacterial community with 16 S rRNA gene high-throughput sequencing on anaerobic 0, 20, and 40 d and aerobic 20 d. (a) Alpha-diversity: The OTUs number, Chao1 and Shannon indexes of the microbial community, (b) beta-diversity: PCoA (Principal coordinates analysis), (c) Top 10 bacterial phyla, and (d) Functional bacterial associated with Hg methylation and demethylation.

3.5. Modelling of Hg transformation under alternative anoxic and oxic conditions

Based on the presented sections, inter-transformation occurred among the available (F0, F1, F2, F3) and relatively immobile (F4, F5, F6, and F7) Hg fractions with the alternation of redox conditions. During anoxic incubation, the enhanced immobilization of available Hg can be roughly classified into three processes: (i) increased sorption on the surface of soil minerals with increasing pH, (ii) complexation by humic acid, larger molecular weight OM, and precipitation as HgS; and (iii) absorption by soil colloids triggered by reductive dissolution of Fe–Mn oxides. During oxic incubation, the increased re-mobilization of Hg with O_2 introduction could be divided into two processes: (i) desorption from the surface of soil minerals with a decrease in pH; (ii) release via the oxidative decomposition of humic acid, stable OM, and oxidative dissolution of HgS, and (iii) release via the re-precipitation of colloids with newly formed Fe oxides.

To quantitatively assess the reaction rates of various processes, this study established a kinetic model based on the aforementioned results, and the detailed setup procedures were as follows: First, a portion of soil OM able to bind metals can be considered reactive OM (ROM), accounting for approximately 45%. In this study, ROM was assumed to be 18% fulvic acid and 82% humic acid (Shi et al., 2013). The Stockholm Humic (SHM) model was selected for the adsorption of metal ions by soil

Table 1
Modeled reactions for Hg transformation under anaerobic and aerobic conditions.

NO.	Reactions				
	Anaerobic condition				
R1	$Hg_{F0123}+Sites_{OM}\leftrightarrow Hg_{F4}$	k_1^+	0.00158	k_1	0.000788
R2	$Hg_{F0123}+Sites_{OM}\leftrightarrow Hg_{F6}$	k_2^+	0.0011	k_2	0.000218
R3	$Hg_{F4}\leftrightarrow Hg_{F6}$	k_3^+	1.08	k_3	0.0183
R4	$Hg_{F0123}+S^{2-}\leftrightarrow Hg_{F6}$	k_4^+	0.401	k_4	0.389
R5	$SO_4^{2-}\leftrightarrow S^{2-}$	k_5^+	0.905	k_5	0.0201
R6	$Hg_{F0123}+Sites_{Oxides\ +\ Clay}\leftrightarrow Hg_{F57}$	k_6^+	0.00162	k_6	0.00412
R7	$Sites_{Increase}\leftrightarrow\triangle Sites$	k_7^+	1.71	k_7	0.501
R8	$Hg_{F0123}\leftrightarrow MeHg$	k_8^+	0.501	k_8	0.102
	Aerobic condition				
R9	$Hg_{F4}\leftrightarrow Hg_{F0123}+Sites_{OM}$	k_9^+	6.72	k_9	0.000319
R10	$Hg_{F6}\leftrightarrow Hg_{F0123}+Sites_{OM}$	k_{10}^+	9.08	k_{10}	0.0353
R11	$Hg_{F6}\leftrightarrow Hg_{F4}$	k_{11}^+	0.508	k_{11}	0.296
R12	$Hg_{F57}\leftrightarrow Hg_{F0123}+Sites_{Oxides\ +\ Clay}$	k_{12}^+	0.000572	k_{12}	0.000103
R13	$Hg_{F6}\leftrightarrow Hg_{F0123}+S^{2-}$	k_{13}^+	58	k_{13}	0.00131
R14	$\triangle Sites\leftrightarrow Sites_{Decrease}$	k_{14}^+	0.00014	k_{14}	0.963
R15	$S^{2-}\rightarrow SO_4^{2-}$	k_{15}^+	0.000239	k_{15}	3.11
R16	$MeHg\leftrightarrow Hg_{F0123}$	k_{16}^+	0.137	k_{16}	0.0045

F0123 = F0+F1+F2+F3 , F57 = F5+F7

organic matter (Gustafsson and Berggren Kleja, 2005). In the SHM model, the adsorption site density of humic acid and fulvic acid was 5.33 mol kg^{-1} and 7.02 mol kg^{-1}, respectively. Therefore, $Sites_{OM}$ can be calculated using Equation (1).

$$Sites_{OM} = TOC * 0.82 * 5.33 + TOC * 0.18 * 7.02 \quad (1)$$

Second, the retention of available Hg by oxides and soil colloids is an important process. Sites of clay and oxides can be calculated by the surface site concentration (*Hs*), and sites of organic matter and can be expressed as Equation (2).

$$Sites_{oxides+clay} = Hs - Sites_{OM} \quad (2)$$

Third, Hg(II) tends to form covalent bonds with reduced sulfur and has a very high affinity for reduced forms of organic sulfur and inorganic sulfides. Sulfide was generated via the biological reduction of SO_4^{2-} (Reaction 5, Table 1).

During the establishment of the model, F0, F1, F2, and F3 were combined (F0123) due to their high availability; F5 was included in F7 (F57) due to its low content; and F7 might contain crystalline Fe–Mn oxide-bound Hg. The reactions (R1-R16) used in the kinetic model and the best-fit values of the rate constants are presented in Table 1. The simulation of the Hg fractions (Fig. 5) matched the experimental data well during the entire incubation period. The fractions increased in the order F6 > F4 > F57 > F0123. The k^+ and k^- represent the rate constants of the forward and backward reactions, respectively. During the anoxic stage, except for R6, k^+ was greater than k^-, indicating forward reactions. k_1^+ (0.00158) and k_2^+ (0.0011) were very close, but k_3^+ (1.08) was three orders of magnitude higher than k_1^+ and k_2^+, suggesting that Hg was first immobilized by humic acid (F4) and the larger molecular weight OM (F6) (R1 and R2), and F4 was gradually transformed to F6 (R3). Hg is preferentially associated with higher-molecular-weight fractions of organic matter (humic substances) (Chakraborty et al., 2014; Padalkar et al., 2019). OM with a high molecular weight can form a stable complex or chelate with Hg, thereby reducing the availability of Hg (Wu et al., 2022). In addition, the reduction of SO_4^{2-} (R5 $k_5^+ = 0.905$) reduced the mobility of Hg by forming insoluble HgS (R4 $k_4^+ = 0.401$) and promoted MeHg formation by stimulating Hg methylator activities (R8 $k_8^+ = 0.501$). By contrast, the backward k_6^- (R6) of mobile Hg (F0123) bound to crystalline Fe–Mn oxides and clay minerals (F57) was higher than the forward rate constant k_6^+, suggesting that some of the structurally incorporated Hg (F57) could be released with the reductive dissolution of crystalline Fe oxides and maintain a low Hg concentration in the soil solution. After O_2 was introduced into the system, as shown in Fig. 5, the modelling results showed that the available Hg (F0123) increased with incubation time. The k^+ of Hg released from humic acid, OM with a larger molecular weight, and HgS ($k_9^+ = 6.72$, $k_{10}^+ = 9.08$, $k_{13}^+ = 58$, respectively) were much higher than those of Hg desorption from crystalline Fe–Mn oxides and MeHg demethylation ($k_{12}^+ = 0.000572$, $k_{16}^+ = 0.137$, respectively). The k_{15}^+ of the oxidation from sulfide to sulfate was relatively low ($k_{15}^+ = 0.000239$), which might be ascribed to the production of intermediates or retention by newly formed Fe oxides (Wang et al., 2022). Therefore, the re-mobilization of Hg upon soil oxygenation prevailed with the oxidative decomposition of OM, dissolution of HgS, and desorption from the surfaces of soil minerals induced by the decreasing pH. However, F4 and F6 were still the dominant fractions, consistent the following: OM significantly affects the solubility and availability of Hg (Sysalová et al., 2017). This kinetic model provides a quantitative understanding of the Hg transformation process in complex acidic paddy soil systems under alternating redox conditions.

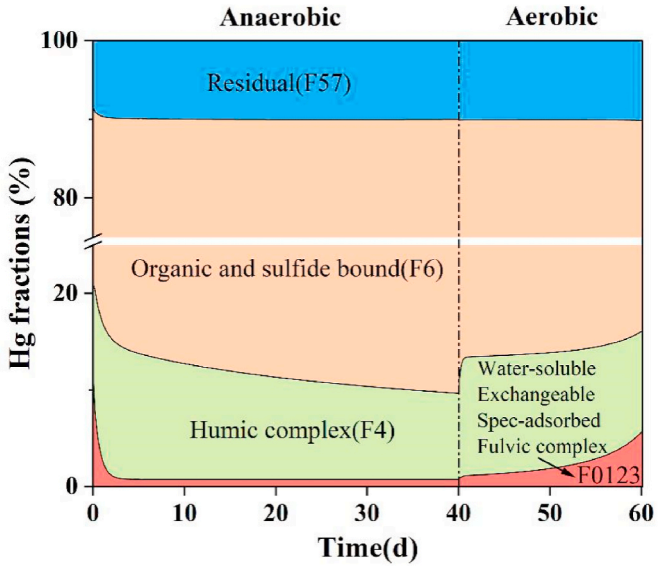


Fig. 5. Transformation of Hg fractions under anaerobic and aerobic conditions according to reactions 1 to 16 (R1-R16) in the paddy soil.

3.6. Implications

External Hg input such as atmospheric deposition, metal mining and smelting, chemical industry emissions and wastewater irrigation has resulted in serious Hg contamination in Southern Asian paddy fields (O'Connor et al., 2019; Tang et al., 2020). Despite the high exogenous Hg input, the fractions of Hg could be quickly equilibrated within 2 d (Fig. 1a), consisting with Fulda et al. (2013). Changes in Hg fractions during the incubation reflected the Hg solution dynamics. Hg was mainly associated with humic acid and large molecular OM, or coprecipitated with sulfur in both flooding and drainage stage (Fig. 1a and Table 1). Since humic acid and large molecular OM have high complex stability potential, the migration of Hg decreased in contaminated soil (Frohne and Rinklebe, 2013). The S-containing functional ligands on these OM are inclined to binding Hg^{2+} intensively (Sysalová et al., 2017). Furthermore, the microbial reduction of SO_4^{2-} (R5, $k_5^+ = 0.905$) also decreased the mobility of Hg by forming insoluble HgS under anoxic conditions (Revis et al., 1989; Devai et al., 2005; Yin et al., 2016). It was reported that HgS was the major Hg form in floodplain soils (85%) in Tennessee, USA (Revis et al., 1989) and in paddy soils (64–81%) in Wanshan Hg Mining District in Guizou, China (Yin et al., 2016). Differently, the combination of Hg with colloidal OM and DOM such as fluvic acid will increase the mobility and availability of Hg (Graham et al., 2012). Although the percentage of F3–Hg as well as F0–Hg, F1–Hg and F2–Hg remained at low levels except that on 0 d (Fig. 1a), they have high mobility and are readily available for rice (Frohne and Rinklebe, 2013). Moreover, the DOM in anoxic conditions was much higher than that under oxic conditions (Fig. 2d), which will promote Hg methylation through boosting the growth of Hg methylated microorganisms such as FeRB and SRB (Graham et al., 2012; Liu et al., 2016). The methylation of Hg under anoxic condition was more conducive than that under oxic conditions (Fig. 1b). Therefore, it will aid in decreasing the accumulation of MeHg in rice grains via reducing flooding period. Besides OM and S, Fe and N also affected the transformation of Hg in the acidic paddy soil (Fig. 2b, c and 2e). Agricultural activities such as fertilizer application and irrigation could introduce anions, cations, and organics into paddy soils, and alter physicochemical characteristics and microorganism communities, resulting in the changes in the transformation and mitigation of Hg (Liu et al., 2016; Liu et al., 2019; Tang et al., 2020). Hg fractions were changing dramatically under alternating redox conditions, so it is essential to assess the transformation of Hg fractions during the rice cultivation. However, the volatility and methylation of Hg, in comparison to other classic heavy metals, have caused attempts to study its dynamics and potential availability very challenging. In this study, according to the key biogeochemical reactions of Hg in paddy soil under the alternating redox conditions, we have established a new kinetic model to quantify the various transformation reactions of Hg fractions.

Based on modified SEP procedures and key elementary reactions for soil biogeochemical processes of heavy metals, similar models have been established to predict the transformation and mitigation of these metals such as cadmium and chromium (Yang et al., 2021; Zhang et al., 2023). It has been demonstrated that all the model-predicted available heavy metals had stronger correlations with their contents in rice grain than those of chemical extraction methods. The kinetic model will be feasible and is expected to predict the availability of Hg in paddy soils in future work.

4. Conclusion

The kinetics of exogenous Hg transformation in an acidic paddy soil was investigated under alternating anoxic and oxic conditions. The concentration of available Hg (F0+F1+F2+F3) decreased in the initial anoxic stage due to binding on humic acid, larger molecular OM or co-precipitation with sulfur, whereas it increased slightly in the later anoxic stage due to the OM degradation and HgS oxidation. The MeHg content reached highest on 20 d of anoxic stage and decreased in the late anoxic stage (20–40 d). In the oxic stage, the content of available Hg, especially water-soluble Hg (F0), kept increasing due to the degradation of OM under oxidizing conditions, while the MeHg production from methylation decreased. The microbes *Firmicutes*, *Proteobacteria*, and *Bacteroidetes* at the phylum level and SRB microorganisms, *Desulfovibrio* and *Desulfomicrobium* at the genus level mediated Hg methylation, whereas *Clostridium_sensu_stricto_1* and *Clostridium_sensu_stricto_12* at the genus level mediated MeHg demethylation under the oxic condition. The kinetic model could be used to predict Hg transformation and migration in flooding-drainage alternated acidic paddy soils. These findings provided new insights into exogenous Hg transformation in periodically flooded acidic paddy soils.

Author statement

Yufan Zhang: Formal analysis, Investigation, Visualization, Methodology, Writing - original draft. **Xiangqin Wang:** Supervision, Visualization, Data Curation, Writing – Review & editing. **Yang Yang**, Conceptualization, Supervision, Writing – Review & editing. **Yingmei Huang:** Formal analysis, Investigation, Validation. **Xiaomin Li:** Visualization, Data Curation, Writing-Review & editing. **Shiwen Hu:** Conceptualization, Supervision, Writing – Review & editing. **Kexue Liu:** Conceptualization, Supervision, Writing – Review & editing. **Yan Pang:** Supervision, Visualization, Validation. **Tongxu Liu:** Investigation, Project administration, Validation. **Fangbai Li:** Conceptualization, Supervision, Writing – Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2023.121335.

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